



University of Xproce
Department of Computer Science

Computer Science Report Template and Report Writing Guide

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A report submitted in partial fulfilment of the requirements of
the Hassan II University
Artificial Intelligence and Computer Science

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Declaration

I, Firstname(s) Lastname, of the Department of Computer Science, University of Reading, confirm that this is my own work and figures, tables, equations, code snippets, artworks, and illustrations in this report are original and have not been taken from any other person's work, except where the works of others have been explicitly acknowledged, quoted, and referenced. I understand that if failing to do so will be considered a case of plagiarism. Plagiarism is a form of academic misconduct and will be penalised accordingly.

I give consent to a copy of my report being shared with future students as an exemplar.

I give consent for my work to be made available more widely to members of UoR and public with interest in teaching, learning and research.

Firstname(s) Lastname
May 27, 2026

Abstract

Guidance on abstract writing: An abstract is a summary of a report in a single paragraph up to a maximum of 250 words. An abstract should be self-contained, and it should not refer to sections, figures, tables, equations, or references. An abstract typically consists of sentences describing the following four parts: (1) introduction (background and purpose of the project), (2) methods, (3) results and analysis, and (4) conclusions. The distribution of these four parts of the abstract should reflect the relative proportion of these parts in the report itself. An abstract starts with a few sentences describing the project's general field, comprehensive background and context, the main purpose of the project; and the problem statement. A few sentences describe the methods, experiments, and implementation of the project. A few sentences describe the main results achieved and their significance. The final part of the abstract describes the conclusions and the implications of the results to the relevant field.

Keywords: A minimum/maximum of four to ten keywords/keyphrases, separated by commas.

Acknowledgements

An acknowledgements section is optional. You may like to acknowledge the support and help of your supervisor(s), friends, or any other person(s), department(s), institute(s), etc. If you have been provided specific facility from department/school acknowledged so.

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List of Abbreviations

SMPCS School of Mathematical, Physical and Computational Sciences

General Introduction

Guidance on introduction chapter writing: Introductions are written in the following parts:

- A brief description of the investigated problem.
- A summary of the scope and context of the project, i.e., what is the background of the topic/problem/application/system/algorithm/experiment/research question/hypothesis/etc. under investigation/implementation/development [whichever is applicable to your project].
- The aims and objectives of the project.
- A description of the problem and the methodological approach adopted to solve the problem.
- A summary of the most significant outcomes and their interpretations: Describe clearly what you have done/created/achieved and what the major results and their implications are.
- Organization of the report.

Two Pages long and use only a single **font** for the body text. We recommend using a clean and electronic document friendly font like **Arial** or **Calibri** for MS-word (If you create a report in MS word). If you use this template, DO NOT ALTER the template's default font "amsfont default computer modern". The default L^AT_EX font "computer modern" is also acceptable.

The recommended body text **font size** is minimum **11pt** and minimum one-half line spacing. The recommended figure/table caption font size is minimum 10pt. The footnote¹ font size is minimum 8pt. DO NOT ALTER the font setting of this template.

Organization of the report

Describe the outline of the rest of the report here. Let the reader know what to expect ahead in the report. Describe how you have organized your report.

Example: how to refer a chapter, section, subsection. This report is organised into seven chapters. Chapter 2 details the literature review of this project. In Section 3...

Note: Take care of the word like "Chapter," "Section," "Figure" etc. before the L^AT_EX command `\ref{}`. Otherwise, a sentence will be confusing. For example, In 2 literature review is described. In this sentence, the word "Chapter" is missing. Therefore, a reader would not know whether 2 is for a Chapter or a Section or a Figure. For more information on **automated tools** to assist in this work, see Section 3.7.3.

¹Example footnote: footnotes are useful for adding external sources such as links as well as extra information on a topic or word or sentence. Use command `\footnote{...}` next to a word to generate a footnote in L^AT_EX.

Chapter 1

Introduction

This chapter provides the first picture of the project. It introduces the host organization, describes the problem being investigated, outlines the aims and objectives, and summarises the solution approach and project management plan.

1.1 Host Organization

1.1.1 Presentation: Name and Logo

1.1.2 Sector and Activities

1.1.3 Missions

1.1.4 Organisational Chart

1.2 Project Presentation

1.2.1 Background

Describe to a reader the context of your project. That is, what is your project and what is its motivation. Briefly explain the major theories, applications, and/or products/systems/algorithms whichever is relevant to your project.

Caution: Do not say you chose this project because of your interest, or your supervisor proposed it, or you were assigned it as your final year project. This may all be true, but it is not meant to be written here.

1.2.2 Problem Statement

This section describes the investigated problem in detail. You can also have a separate chapter on “Problem Articulation.” For some projects, you may have a section like “Research Question(s)” or “Research Hypothesis” instead of a section on “Problem Statement.”

1.2.3 Aims and Objectives

Describe the “aims and objectives” of your project.

Aims: The aims tell a reader what you want/hope to achieve at the end of the project. The aims define your intent and purpose in general terms.

Objectives: The objectives are a set of tasks you would perform in order to achieve the defined aims. The objective statements must be specific and measurable through the results and outcomes of the project.

1.2.4 Solution Approach

Briefly describe the solution approach and the methodology applied in solving the set aims and objectives.

Depending on the project, you may like to alter the heading of this section. Check with your supervisor. Also check what subsections or other sections can be added or removed from this template.

1.3 Project Management

This chronological outline provides a structured overview of the project's execution steps, as illustrated in the accompanying Gantt diagram. It delineates the planned sequence of activities and milestones, offering a roadmap for the project's progression from initiation to completion.

1.4 Summary

In summary, this chapter has introduced the host organization and established the context in which this project is situated. It described the problem statement, the aims and objectives, and the solution approach adopted. The project management plan was also outlined. Chapter [2](#) presents a review of the existing literature relevant to this project.

Chapter 2

Literature Review

This chapter reviews the existing literature relevant to the investigated problem. It situates the project within the current state of the art, identifies gaps in existing knowledge, and demonstrates the significance and motivation of this work. The review is organised as follows: Section 2.1 provides guidance on structuring a literature review; Section 2.2 discusses critical analysis and synthesis; Section 2.3 covers referencing and citation conventions; and Section 2.4 addresses plagiarism avoidance.

A literature review chapter might contain the following:

1. A review of the state-of-the-art (include theories and solutions) of the field of research.
2. A description of the project in the context of existing literature and products/systems.
3. An analysis of how the review is relevant to the intended application/system/problem.
4. A critique of existing work compared with the intended work.

Note that your literature review should demonstrate the significance of the project.

2.1 Organising Your Literature Review

A literature review is not a list of paper summaries. It must be organised around ideas, themes, or arguments — not around individual papers. There are two main organisational strategies:

- **Thematic organisation (recommended):** Group sources by topic, concept, or research theme. For example, one section may review methods for feature extraction, another for classification, and another for evaluation benchmarks. This approach shows the reader the landscape of ideas and how they relate to each other.
- **Chronological organisation:** Arrange sources from earliest to most recent to show how the field has evolved over time. This is useful when the development of a concept over time is central to your argument.

Regardless of the structure chosen, each section of your literature review should:

- Open with a sentence that states what the section is about.
- Discuss multiple sources together and compare them, rather than describing each paper in isolation.
- End with a sentence that connects the reviewed material to your own project.

2.2 Synthesis and Critical Analysis

The most important skill in a literature review is synthesis: combining information from multiple sources to build a coherent argument, rather than simply summarising each paper in turn.

- **Summarising** (avoid as a primary mode): “Smith (2020) proposed method X. Jones (2021) proposed method Y. Lee (2022) proposed method Z.”
- **Synthesising** (aim for this): “Several methods have been proposed for X, including approach A (Lamport, 1994), approach B (Kottwitz, 2021), and approach C. While A and B share the assumption that..., C differs by... However, none of these approaches address the problem of..., which is the gap this project targets.”

Critical analysis means evaluating the strengths and weaknesses of existing work — not just reporting what was done. For each group of sources, ask:

- What are the key assumptions made, and are they justified?
- What are the limitations of the proposed methods?
- How do the results compare across different studies?
- What remains unresolved or contested in the literature?

2.3 Citation and Referencing Guide

The references in a report relate your content with the relevant sources, papers, and the works of others. To include references in a report, we *cite* them in the text. In $\LaTeX$, the most convenient way is to use BibTeX, which takes references in a particular format [see `references.bib` file of this template] and formats them in the required style (APA, Harvard, IEEE, etc.) with the help of appropriate packages.

Take note of the commands `\cite{}` and `\citep{}`. The command `\cite{}` produces “Author et al. (2019)” style for Harvard/APA. The command `\citep{}` produces “(Author et al., 2019).” Depending on how you construct a sentence, use them appropriately. This Department recommends the **Harvard style** of referencing.

- Kottwitz (2021) has written a comprehensive guide on writing in $\LaTeX$ [Example of `\cite{}`].
- If $\LaTeX$ is used efficiently and effectively, it helps in writing a very high-quality project report (Lamport, 1994) [Example of `\citep{}`].
- A detailed APA, Harvard, and Chicago referencing style guide are available in (University of Reading, 2023b).

This is an example of a numbered list with in-text citations:

1. Kottwitz (2021) has written a comprehensive guide on writing in $\LaTeX$.
2. If $\LaTeX$ is used efficiently and effectively, it helps in writing a very high-quality project report (Lamport, 1994).

2.3.1 Reference Resources

You can find additional referencing resources from the University Library:

- <https://libguides.reading.ac.uk/computer-science>
- <https://libguides.reading.ac.uk/citing-references/citationexamples>

2.3.2 Changing Bibliography Styles

While this report uses name-and-date formatting in the Harvard style, you may also wish to use a numbered style such as IEEE. To enable this, edit the `CS_report.sty` file. Uncomment the 2 lines of Harvard settings under Bibliography/References settings, and enable the IEEE style:

```
% IEEE, Numbered Style
\usepackage[numbers]{natbib}
\bibliographystyle{IEEEtran}
```

Note that when making this change, you will need to modify the way in which you refer to authors by name using `\citeauthor{}`.

2.4 Avoiding Unintentional Plagiarism

Using other sources, ideas, and material always carries a risk of unintentional plagiarism.

MUST: Read the university guidelines on the definition of plagiarism as well as the guidelines on how to avoid it ([University of Reading, 2023a](#)).

2.5 Critique of the Review

Describe your main findings and evaluation of the literature. What are the key gaps, contradictions, or open problems identified through your review? How does this project address one or more of these gaps?

2.6 Summary

This chapter reviewed the existing literature relevant to [state the topic of your project]. The review identified that [summarise the key themes covered]. It was found that [state the main gap or limitation in existing work] which motivates the approach adopted in this project. The following chapter, Chapter 3, describes the methodology used to address this gap.

Chapter 3

Methodology

This chapter describes the methodology adopted to address the research objectives stated in Chapter 1. It outlines the overall report structure, describes the experimental design, presents the implementation details, and addresses the ethical considerations relevant to this project.

3.1 Report Structure and Methodology Type

A project report's structure can follow different paradigms depending on the type of project. Table 3.1 summarises the general report structure used in this template.

Table 3.1: Undergraduate report template structure

Frontmatter	Title Page
	Abstract
	Acknowledgements
	Table of Contents
	List of Figures
	List of Tables
	List of Abbreviations
Main text	Chapter 1 Introduction
	Chapter 2 Literature Review
	Chapter 3 Methodology
	Chapter 4 Results
	Chapter 5 Discussion and Analysis
	Chapter 6 Conclusions and Future Work
End matter	Chapter 7 Reflection
	References
	Appendices (Optional)
Index (Optional)	

3.1.1 Example of an Algorithm Analysis Main Text Structure

Some projects involve the implementation of a state-of-the-art algorithm and its performance analysis and comparison with other algorithms. In that case, the suggestion in Table 3.2 may

suit you best.

Table 3.2: Example of an algorithm analysis type report structure

Chapter 1	Introduction	
Chapter 2	Literature Review	
Chapter 3	Methodology	Algorithms descriptions Implementations Experiments design
Chapter 4	Results	
Chapter 5	Discussion and Analysis	
Chapter 6	Conclusion and Future Work	
Chapter 7	Reflection	

3.1.2 Example of an Application Type Main Text Structure

If you are applying some algorithms/tools/technologies on some problems/datasets/etc., you may use the structure in Table 3.3.

Table 3.3: Example of an application type report structure

Chapter 1	Introduction	
Chapter 2	Literature Review	
Chapter 3	Methodology	Problems (tasks) descriptions Algorithms/tools/technologies/etc. descriptions Implementations Experiments design and setup
Chapter 4	Results	
Chapter 5	Discussion and Analysis	
Chapter 6	Conclusion and Future Work	
Chapter 7	Reflection	

3.1.3 Example of a Science Lab-Type Main Text Structure

If you are doing a science lab experiment type of project, you may use the methodology section suggested in Table 3.4. In this kind of project, you may refer to the “Methodology” chapter as “Materials and Methods.”

3.2 Ethical Considerations

This section addresses the ethical aspects of your project. Consider all dimensions that apply to your work:

- **Informed Consent:** If data was collected from participants, detail the process for obtaining consent and ensuring participants understand their rights.

Table 3.4: Example of a science lab experiment-type report structure

Chapter 1	Introduction	
Chapter 2	Literature Review	
Chapter 3	Materials and Methods	Problems (tasks) description Materials Procedures Implementations Experiment set-up
Chapter 4	Results	
Chapter 5	Discussion and Analysis	
Chapter 6	Conclusion and Future Work	
Chapter 7	Reflection	

- **Confidentiality and Privacy:** Explain measures taken to protect participants' data and maintain confidentiality. Discuss how data is stored, who has access, and how anonymity is preserved.
- **Risk Assessment:** Identify potential risks to participants and outline strategies to minimise them.
- **Vulnerable Populations:** If applicable, address how you will protect vulnerable groups (e.g., children, elderly, or marginalised communities) involved in your project.
- **Research Integrity:** Highlight your commitment to honesty and transparency. Discuss how you will avoid plagiarism, fabrication, and falsification of data.
- **Compliance with Regulations:** Mention the relevant ethical guidelines and regulations your project adheres to.
- **Impact on Society:** Reflect on the broader implications of your project. Discuss how outcomes may affect communities, stakeholders, or the environment, and how you plan to address any potential negative consequences.
- **Feedback Mechanisms:** Describe how you incorporate feedback from participants and stakeholders to improve the ethical conduct of the project throughout its duration.

3.3 Example of an Equation in $\LaTeX$

Eq. (3.1) is an example of an equation in $\LaTeX$. In Eq. (3.1), s is the mean of elements $x_i \in \mathbf{x}$:

$$s = \frac{1}{N} \sum_{i=1}^N x_i. \quad (3.1)$$

Have you noticed that all the variables of the equation are defined using the **in-text** maths command $\$$, and Eq. (3.1) is treated as part of the sentence with proper punctuation? Always treat an equation or expression as part of a sentence.

3.4 Example of a Figure in $\text{\LaTeX}$

Figure 3.1 is an example of a figure in $\text{\LaTeX}$. For more details, check the link: wiki-books.org/wiki/LaTeX/Floats,_Figures_and_Captions.

Keep your artwork (graphics, figures, illustrations) clean and readable. At least 300dpi is a good resolution for PNG format artwork. However, an SVG format artwork saved as a PDF will produce the best quality graphics. One useful tool is “Flow algorithm software”: flowgorithm.org.

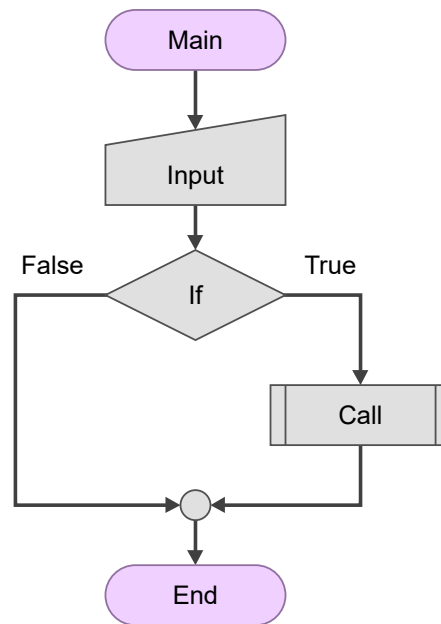


Figure 3.1: Example figure in $\text{\LaTeX}$.

3.5 Example of an Algorithm in $\LaTeX$

Algorithm 1 is a good example of an algorithm in $\LaTeX$.

Algorithm 1 Example caption: sum of all even numbers

Input: $\mathbf{x} = x_1, x_2, \dots, x_N$

Output: *EvenSum* (Sum of even numbers in $\mathbf{x}$)

```

1: function EvenSummation( $\mathbf{x}$ )
2:   EvenSum  $\leftarrow$  0
3:    $N \leftarrow \text{length}(\mathbf{x})$ 
4:   for  $i \leftarrow 1$  to  $N$  do
5:     if  $x_i \bmod 2 == 0$  then                                 $\triangleright$  Check whether a number is even.
6:       EvenSum  $\leftarrow$  EvenSum +  $x_i$ 
7:     end if
8:   end for
9:   return EvenSum
10: end function

```

3.6 Example of a Code Snippet in $\LaTeX$

Code Listing 3.1 is a good example of including a code snippet in a report. While using code snippets, take care of the following:

- Do not paste your entire implementation. Add code snippets only.
- The algorithm shown in Algorithm 1 is usually preferred over code snippets in a technical/scientific report.
- Make sure the entire code snippet or algorithm stays on a single page and does not overflow to another page.

Here are three examples of code snippets for Python, Java, and C++ illustrated in Listings 3.1, 3.2, and 3.3 respectively.

```

1 import numpy as np
2
3 x = [0, 1, 2, 3, 4, 5]          # assign values to an array
4 evenSum = evenSummation(x)     # call a function
5
6 def evenSummation(x):
7     evenSum = 0
8     n = len(x)
9     for i in range(n):
10         if np.mod(x[i], 2) == 0: # check if a number is even?
11             evenSum = evenSum + x[i]
12     return evenSum

```

Listing 3.1: Code snippet in $\LaTeX$ — Python example

```

1 public class EvenSum {
2     public static int evenSummation(int[] x) {
3         int evenSum = 0;
4         int n = x.length;
5         for (int i = 0; i < n; i++) {
6             if (x[i] % 2 == 0) { // check if a number is even?
7                 evenSum = evenSum + x[i];
8             }
9         }
10        return evenSum;
11    }
12    public static void main(String[] args) {
13        int[] x = {0, 1, 2, 3, 4, 5}; // assign values to an array
14        int evenSum = evenSummation(x);
15        System.out.println(evenSum);
16    }
17 }

```

Listing 3.2: Code snippet in L^AT_EX — Java example

```

1 int evenSummation(int x[]) {
2     int evenSum = 0;
3     int n = sizeof(x);
4     for (int i = 0; i < n; i++) {
5         if (x[i] % 2 == 0) { // check if a number is even?
6             evenSum = evenSum + x[i];
7         }
8     }
9     return evenSum;
10 }
11
12 int main() {
13     int x[] = {0, 1, 2, 3, 4, 5}; // assign values to an array
14     int evenSum = evenSummation(x);
15     cout << evenSum;
16     return 0;
17 }

```

Listing 3.3: Code snippet in L^AT_EX — C/C++ example

3.7 In-Text Citation Style Guide

3.7.1 Placement and First Reference

Make sure that whenever you refer to an equation, table, figure, algorithm, or listing for the first time, it also appears on the same page or the following page(s). Always refer to every equation, table, and figure in the text. Do not leave them without an **in-text citation**.

3.7.2 Notation Conventions

Write **Eq.** with an uppercase “E” for an equation before its number (use `\eqref{.}`). Use “Table” to refer to a table, “Figure” to refer to a figure, “Algorithm” to refer to an algorithm, and “Listing” to refer to code snippets. Note that we do not use the articles “a,” “an,” or “the” before Eq., Figure, Table, or Listing.

For example: “A report structure is shown in **the** Table 3.1” should be written as “A report structure is shown **in** Table 3.1.”

3.7.3 Tools for In-Text Referencing

This is enabled by a combination of `\label{}` and `\ref{}` commands. For example: `\section{My Section}\label{sec:my}` creates a label that can be referenced with `\ref{sec:my}`.

In-text equation references use: `Eq.~\eqref{eq:eq_example}`. Alternatively, use the `cleverref` system: `\Cref{eq:eq_example}` yields [Equation \(3.1\)](#). Lowercase `\cref{fig:chart_a}` yields [fig. 3.1](#).

3.8 Summary

This chapter described the methodology adopted in this project. The overall report structure was introduced and three project type templates were presented to guide the reader in adapting it to their needs. The ethical considerations relevant to this project were discussed. The chapter also provided practical $\text{\LaTeX}$ guidance on equations, figures, algorithms, code listings, and in-text referencing conventions. Chapter [4](#) presents the results obtained by applying the methodology described here.

Chapter 4

Results

This chapter presents the results obtained by applying the methodology described in Chapter 3. The findings are arranged in a logical sequence, progressing from overall algorithm performance to data-driven findings, and concluding with the ablation study. Each result is directly linked to a specific objective or research question stated in Chapter 1.

The results chapter tells a reader about your findings based on the methodology you have used to solve the investigated problem. For example:

- If your project aims to implement an algorithm for its analysis, the results may be the performance of the algorithm obtained using a relevant experiment design.
- If your project aims to solve some problems/research questions over a collected dataset, the results may be the findings obtained using the applied tools/algorithms/etc.

4.1 How to Present Your Results

Arrange your results and findings in a logical sequence. Use a bottom-up approach by first presenting raw or processed outputs, then progressively building toward interpretation. Each result should directly correspond to an objective or research question stated in your introduction.

When presenting your results, keep the following principles in mind:

- **Be objective:** Report what you observed without over-interpreting. Save deeper analysis for the Discussion chapter.
- **Be precise:** Use exact numerical values, units, and statistical measures (e.g., accuracy, F1-score, RMSE) where applicable.
- **Be complete:** Report both positive and negative findings. Unexpected or poor results are equally important and should not be omitted.
- **Be consistent:** Use the same notation and terminology throughout to ensure clarity.

4.2 Algorithm Performance Results

If your project involves the implementation or evaluation of an algorithm, report its measured performance directly. Table 4.1 presents the quantitative results obtained across the evaluated configurations.

Figure 4.1 provides a visual comparison of the obtained results across all evaluated configurations.

Table 4.1: Performance results of the proposed algorithm.

Model / Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Baseline	—	—	—	—
Proposed Algorithm	—	—	—	—
Variant A	—	—	—	—
Variant B	—	—	—	—

Figure 4.1: Comparison of algorithm performance across evaluated configurations.

4.3 Data-Driven Findings

If your project addresses research questions over a collected dataset, present the findings obtained from applying your analytical methods. Each finding should directly answer a specific research question or objective.

Research Question 1: [State RQ1 here] [Describe the finding that answers RQ1. Reference the relevant table or figure.]

Research Question 2: [State RQ2 here] [Describe the finding that answers RQ2. Reference the relevant table or figure.]

Research Question 3: [State RQ3 here] [Describe the finding that answers RQ3. Reference the relevant table or figure.]

Use the most appropriate visualisation to support each finding:

- **Bar/pie charts** — categorical distributions or group comparisons.
- **Line plots** — trends over time or across parameter values.
- **Scatter plots** — relationships between continuous variables.
- **Heatmaps / confusion matrices** — classification performance or correlation structure.
- **Box plots** — statistical spread and outlier detection.

Figure 4.2: [Caption describing the main finding illustrated by this figure.]

4.4 Ablation Study

To assess the contribution of each component of the proposed approach, an ablation study was conducted by systematically removing or replacing individual components and measuring the impact on overall performance. Table 4.2 reports the results obtained for each ablated variant compared to the full model.

Figure 4.3 illustrates the performance drop associated with the removal of each individual component relative to the full model.

Table 4.2: Ablation study results: performance of each model variant.

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Full Model (Proposed)	—	—	—	—
w/o Component A	—	—	—	—
w/o Component B	—	—	—	—
w/o Component C	—	—	—	—
Baseline (no components)	—	—	—	—

Figure 4.3: Performance drop per removed component relative to the full model.

4.5 Summary

Table 4.3 provides a concise overview of the key results obtained, mapped to each project objective or research question, including the ablation study findings.

Table 4.3: Summary of results mapped to project objectives.

Objective / RQ	Method / Tool Applied	Key Result
Objective 1	[Method used]	[Result obtained]
Objective 2	[Method used]	[Result obtained]
Objective 3	[Method used]	[Result obtained]
Ablation Study	Component removal	[Key finding]

The results presented in this chapter are further interpreted and contextualised in Chapter 5, where they are discussed in relation to the existing literature and the initial research objectives.

Chapter 5

Discussion and Analysis

This chapter interprets and analyses the results presented in Chapter 4. It explains what the findings mean in the context of the research objectives, situates them within the existing literature, and evaluates their significance. The limitations of the project are also discussed, along with directions for future improvement.

Depending on the type of project you are doing, this chapter can be merged with the “Results” chapter as “Results and Discussion” as suggested by your supervisor. In the case of software development and standalone applications, describe the significance of the obtained results and the overall performance of the system.

5.1 Interpretation of Results

The Discussion and Analysis chapter evaluates and interprets the results presented in Chapter 4. Unlike the Results chapter, which reports findings objectively, this chapter explains *what the results mean, why they occurred, and how they relate to the research objectives and existing literature*.

When interpreting your results, consider the following:

- **Relate back to objectives:** For each result, explicitly state whether it supports, partially supports, or contradicts the corresponding research objective or hypothesis.
- **Compare with the literature:** Situate your findings within the broader body of knowledge. Do your results agree with or diverge from prior work? If they diverge, propose a reasoned explanation.
- **Explain unexpected outcomes:** If any result was surprising or inconsistent with expectations, discuss possible causes such as data quality, model assumptions, or experimental constraints.
- **Avoid repetition:** Do not simply restate the numbers from the Results chapter. Instead, provide analytical commentary that adds new insight.

5.1.1 Interpretation for Algorithm-Based Projects

If your project involves algorithm implementation or evaluation, interpret the measured performance in context. For example:

- Does the achieved accuracy meet the threshold required for the application domain?

- How does the proposed algorithm compare to the baseline in terms of performance, complexity, and scalability?
- Are the gains in accuracy justified given any trade-offs in computational cost or memory usage?
- Do the results generalise across different configurations or datasets, or are they specific to the experimental conditions?

5.1.2 Interpretation for Data-Driven Projects

If your project addresses research questions over a collected dataset, interpret each finding in relation to its corresponding research question:

- Does the finding provide a clear and direct answer to the research question, or is the evidence only partial?
- Are there patterns, correlations, or trends that were not anticipated in the problem formulation?
- How do the findings contribute to the understanding of the investigated problem beyond what was already known from the literature?

5.1.3 Analysis of Ablation Study

The ablation study results presented in Table 4.2 provide insight into the individual contribution of each component to the overall performance of the proposed approach. The interpretation below explains what each component's removal reveals about the design choices made.

- **Component A:** Removing Component A resulted in the largest performance drop, suggesting it plays the most critical role in [describe its function, e.g., feature extraction, regularisation, etc.]. This aligns with [cite relevant work] which similarly reported that [related finding].
- **Component B:** The removal of Component B led to a moderate decline in performance, indicating that while it contributes meaningfully, the model retains partial capability without it. This suggests [interpretation].
- **Component C:** Component C showed the smallest individual impact on performance. However, its combined absence with other components may indicate an interdependency that warrants further investigation.

Overall, the ablation study confirms that all components contribute positively to the proposed approach, and that the full model configuration achieves the best trade-off between [metric 1] and [metric 2].

5.2 Significance of the Findings

In this section, discuss the significance of your results and key findings in order to enhance the reader's understanding of the investigated problem. Significance can be assessed at multiple levels:

- **Technical significance:** Does the work introduce a novel method, improve an existing approach, or demonstrate a result that advances the technical state of the art? For example, a meaningful improvement in classification accuracy or a reduction in computational complexity may constitute technical significance.
- **Practical significance:** Can the findings be applied in a real-world context? Discuss whether the results are actionable, deployable, or useful for practitioners in the relevant domain.
- **Scientific significance:** Do the findings confirm, challenge, or extend existing theories or models in the field? Even negative results can be scientifically significant if they rule out an assumed relationship or expose a limitation in a widely used method.

Where applicable, explicitly state how the findings address the gap in the literature identified in Chapter 2, and what new knowledge this project contributes.

5.3 Limitations

Every project operates under constraints that may affect the validity, generalisability, or completeness of its findings. Acknowledging limitations transparently demonstrates critical thinking and scientific maturity. Discuss the key limitations of your project under the following aspects:

- **Data limitations:** Were there constraints on the size, diversity, or quality of the dataset? For instance, a small or imbalanced dataset may limit the generalisability of the results to broader populations or real-world conditions.
- **Methodological limitations:** Are there assumptions underlying your chosen methods that may not hold in all cases? For example, certain algorithms assume linearity, stationarity, or independence of features.
- **Scope limitations:** Were certain aspects of the problem left unexplored due to time, resource, or access constraints? Be explicit about what is outside the scope of this work and why.
- **Evaluation limitations:** Are the chosen evaluation metrics fully representative of real-world performance? For instance, accuracy alone may be misleading in the presence of class imbalance.
- **Reproducibility:** Are there factors in your experimental setup (e.g., randomness, proprietary tools, hardware dependency) that may affect the reproducibility of the results?

Where relevant, suggest potential improvements or directions that could address these limitations in future work. This naturally transitions into the recommendations discussed in Chapter 6.

5.4 Summary

This chapter provided an analytical interpretation of the results presented in Chapter 4. The key findings were examined in relation to the project objectives, situated within the existing literature, and assessed for their technical, practical, and scientific significance. The ablation

study was analysed to evaluate the contribution of individual components to the overall system performance. The main limitations of the project were also identified and discussed, along with potential directions for improvement.

Overall, the results *[REPLACE: support / partially support / challenge]* the initial research objectives by demonstrating that *[REPLACE: insert a one or two sentence summary of the core finding here]*. The following chapter presents the conclusions drawn from this work and outlines recommendations for future research.

Chapter 6

Conclusions and Future Work

This chapter brings the report to a close. Section 6.1 summarises the project’s key contributions and findings in relation to the stated aims and objectives. Section 6.2 outlines directions for future work that build on the results and limitations identified in Chapter 5.

6.1 Conclusions

This project investigated *[REPLACE: state the problem that was investigated]*. The main aim was to *[REPLACE: restate the aim]*, with the following objectives: *[REPLACE: briefly list the objectives]*.

To achieve these objectives, *[REPLACE: briefly describe the methods, algorithms, or tools applied and the experimental setup used]*.

The key findings of this project are as follows:

- *[REPLACE: State the finding related to Objective 1 and whether it was achieved.]*
- *[REPLACE: State the finding related to Objective 2 and whether it was achieved.]*
- *[REPLACE: State the finding related to Objective 3 and whether it was achieved.]*

Overall, the central contribution of this project is *[REPLACE: state the most significant achievement or contribution of the work]*. The results demonstrate that *[REPLACE: summarise the most important result and its implication]*.

A good conclusions summary is approximately 300–500 words. Remove this note before submitting.

6.2 Future Work

The limitations identified in Chapter 5 suggest several directions for future investigation. The following areas represent the most promising avenues for extending this work:

- **[REPLACE: Short-term improvement 1]:** *[REPLACE: Describe a concrete, achievable improvement that directly addresses a limitation identified in the Discussion chapter, e.g., increasing the dataset size, testing on additional benchmarks, tuning hyperparameters.]*
- **[REPLACE: Short-term improvement 2]:** *[REPLACE: Describe a second near-term improvement or extension.]*

In the longer term, the following research directions could build on the outcomes of this project:

- **[REPLACE: Long-term direction 1]:** *[REPLACE: Describe a broader research direction that this project opens up, e.g., applying the method to a different domain, integrating it into a larger system, or combining it with a complementary technique.]*
- **[REPLACE: Long-term direction 2]:** *[REPLACE: Describe a second longer-term direction.]*

A good future work summary is approximately 300–500 words. Remove this note before submitting.

6.3 Summary

This chapter concluded the project by summarising the key findings in relation to the stated aims and objectives, and by identifying the main contributions of this work. Directions for future research were also proposed, grounded in the limitations discussed in Chapter 5. Chapter 7 provides a personal reflection on the project experience.

Chapter 7

Reflection

This chapter presents a personal reflection on the project experience. It goes beyond a list of technical skills to examine the decision-making process, the challenges encountered, and the lessons learned throughout the project. Aim for approximately 300–500 words in total.

Knowledge and Skills Developed. *[REPLACE: Describe the most significant knowledge and technical skills you developed during this project. Go beyond listing tools and languages — reflect on what you now understand about the problem domain, research methods, or system design that you did not understand before.]*

Decision-Making and Problem-Solving. *[REPLACE: Reflect on how you approached decision-making during the project. For example: How did you decide which method to use? How did you respond when an approach did not work as expected? What did this teach you about the process of engineering or research?]*

Challenges Encountered. *[REPLACE: Describe the most significant challenges you faced during the project, including any that you were not fully able to overcome. Be honest and specific. Reflect on why these challenges arose and what you learned from attempting to address them.]*

What I Would Do Differently. *[REPLACE: If you were to start this project again with the knowledge you now have, what would you do differently? This could include choices about methodology, tools, time management, or scope. Reflect on any deviations from your original plan and whether they were justified.]*

Impact on Future Development. *[REPLACE: Reflect on how this project experience has shaped your thinking about future work. What new questions has it raised for you? How has it influenced your understanding of the field or your own capabilities as a researcher or developer?]*

The above structure is a guide. You may reorganise the paragraphs to suit your own reflective narrative, as long as you address all four themes: skills developed, decision-making, challenges, and what you would do differently. Remove this note before submitting.

7.1 Summary

This chapter provided a personal reflection on the project experience, covering the skills and knowledge developed, the decision-making approach adopted, the challenges encountered, and the lessons learned. It concludes the main body of this report. The References section follows, listing all sources cited throughout this work.

Note on References vs. Bibliography: The next section is “References,” which will be

automatically generated if you are using the BibT_EX referencing method. Note the difference: a “References” list contains only the sources you have cited in the report, whereas a “Bibliography” also includes sources you read but did not explicitly cite. This template uses “References” only, which is the recommended approach.

References

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URL: <https://libguides.reading.ac.uk/citing-references/referencingstyles>

Appendix A

An Appendix Chapter (Optional)

Some lengthy tables, codes, raw data, length proofs, etc. which are **very important but not essential part** of the project report goes into an Appendix. An appendix is something a reader would consult if he/she needs extra information and a more comprehensive understating of the report. Also, note that you should use one appendix for one idea.

An appendix is optional. If you feel you do not need to include an appendix in your report, avoid including it. Sometime including irrelevant and unnecessary materials in the Appendices may unreasonably increase the total number of pages in your report and distract the reader.

Appendix B

An Appendix Chapter (Optional)

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